

Uncertainty-Gated Safety for Out-of-Distribution Mapless Navigation

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Abstract: In this study, we propose an uncertainty-driven safety mechanism for pre-trained mapless navigation policies that improves robustness under distribution shift without significantly compromising goal-directed behavior. Safe navigation in unstructured environments remains challenging at deployment time, as learned policies can become overconfident and cause collisions when the environment differs from training data. In contrast, non-adaptive safety filters often act too conservatively and can lead to stagnation. Our method gates Control Barrier Function shielding using epistemic uncertainty estimated by an ensemble transition model. The shield is activated only when the current state departs from the training distribution. This selective activation preserves performance of learned policies while introducing conservative corrections only when needed. Extensive evaluation across static cluttered scenarios, procedurally generated out-of-distribution layouts, and real-world deployment shows that our approach consistently reduces unsafe outcomes and reduces the stagnation often observed with continuous shielding.

Keywords: Reinforcement Learning, Mapless Navigation, Uncertainty-Gated Shielding

1 Introduction

The deployment of autonomous mobile robots in unstructured environments requires balancing operational efficiency with collision avoidance. Reinforcement Learning (RL) has become a widely used paradigm for mapless navigation, but policies trained in simulation or limited environment distributions often fail under deployment-time distribution shift. In unseen layouts, learned policies may become overconfident and collide, while conservative safety mechanisms may prevent progress and cause stagnation [1].

A common strategy for improving safety is to modify the training objective through collision penalties or constrained MDP formulations [2, 3, 4]. Although these methods encourage safer behavior in expectation, they remain sensitive to reward or constraint design and provide limited protection when the test environment differs structurally from the training distribution. This is especially problematic in mapless navigation, where the policy must generalize from local observations alone.

Deployment-time shielding provides a complementary solution by correcting unsafe actions online. Control Barrier Functions (CBFs) [5] and Hamilton–Jacobi reachability [6, 7] offer principled safety filters, but their reliance on conservative assumptions can make them overly restrictive in unfamiliar environments. In practice, continuously active shielding may reduce collisions while inducing oscillations, freezing, or complete failure to reach the goal.

Recent learning-based safety filters, including neural barrier functions and verification-aided approaches, reduce the need for exact analytical models [8, 9, 10]. Related work also combines probabilistic enumeration of unsafe regions with CBF-based deployment-time correction [11]. However, these approaches do not explicitly address when safety filters should be activated under out-of-

38 distribution (OOD) conditions, leaving open the problem of balancing robustness and progress under
 39 structural geometric shifts and sim-to-real mismatch.

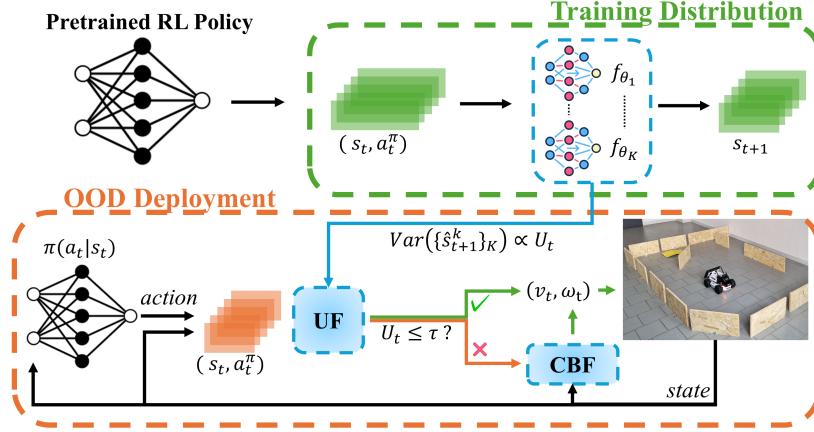


Figure 1: Overview of the uncertainty-gated safety framework. The UF estimates epistemic uncertainty with an ensemble transition model and during deployment activates the CBF shield only under detected distribution shift.

40 We address this problem with an uncertainty-gated safety framework. Our method estimates epis-
 41 temic uncertainty [12] using an ensemble transition model trained on nominal-policy rollouts. At
 42 deployment time, this uncertainty acts as a gate for CBF shielding: the nominal action is preserved in
 43 familiar states, while the CBF correction is activated only when the current state-action pair departs
 44 from the training distribution. This selective intervention reduces unsafe behavior without imposing
 45 the constant conservatism of continuous shielding.

46 Our contributions are: (i) an **uncertainty-based gate** for minimally invasive deployment-time
 47 shielding, compatible with pre-trained navigation policies and action filters; (ii) a **procedural OOD**
 48 **benchmark** for safe mapless navigation under controlled geometric shifts; and (iii) **simulation and**
 49 **real-world validation** showing reduced collisions and less stagnation than continuous shielding.

50 2 Background

51 2.1 Epistemic Uncertainty

52 Uncertainty in learned navigation can be broadly divided into aleatoric uncertainty, due to irreducible
 53 noise, and epistemic uncertainty, due to limited training data. Epistemic uncertainty is particularly
 54 relevant for OOD deployment because it increases when the agent visits state-action regions poorly
 55 covered by the training distribution.

56 In our setting, at time t , the policy proposes $a_t^\pi = \pi(s_t)$. We estimate epistemic uncertainty with an
 57 ensemble of K stochastic one-step transition models $\{f_{\theta_k}\}_{k=1}^K$, trained on data $\mathcal{D} = \{(s_t, a_t^\pi, s_{t+1})\}$
 58 collected under π :

$$\hat{s}_{t+1}^{(k)} \sim f_{\theta_k}(s_t, a_t^\pi), \quad k = 1, \dots, K. \quad (1)$$

59 Let $\mu_{t+1}^{(k)} \in \mathbb{R}^d$ be the predicted mean next state of model k . Ensemble disagreement serves as a
 60 practical proxy for epistemic uncertainty, since diverging predictions among members trained on the
 61 same data indicate that the input (s_t, a_t^π) lies outside the training manifold. We use the score

$$U_t = \frac{1}{d} \sum_{j=1}^d \text{Var}_k(\mu_{t+1,j}^{(k)}) = \frac{1}{d} \text{tr}(\text{Cov}(\{\mu_{t+1}^{(k)}\}_{k=1}^K)). \quad (2)$$

62 Since state variables are standardized, averaging across dimensions prevents any single component
 63 from dominating the uncertainty score.

64 2.2 Control Barrier Functions as Safety Filters

65 CBFs enforce safety at execution time by constraining the control input so that a predefined safe set
 66 remains forward invariant [5]. We denote by x_t the agent state used by the CBF, which is a subset of
 67 the full policy state s_t . The safe set is defined as

$$\mathcal{S} = \{x \mid h(x) \geq 0\}, \quad (3)$$

68 where h is a continuously differentiable barrier function encoding the safety margin between the
 69 robot and perceived obstacles. For a control input a , safety is enforced through the standard CBF
 70 condition

$$\dot{h}(x_t, a) + \alpha h(x_t) \geq 0, \quad (4)$$

71 where $\alpha > 0$ regulates the strength of the correction. Given a nominal policy action a_t^π , the filtered
 72 action is obtained by solving the per-step quadratic program

$$a_t^{\text{safe}} = \arg \min_a \|a - a_t^\pi\|^2 \quad \text{s.t.} \quad \dot{h}(x_t, a) + \alpha h(x_t) \geq 0. \quad (5)$$

73 This projection minimally modifies the policy command while satisfying the CBF constraint associ-
 74 ated with forward invariance under the assumed local model.

75 3 Problem Statement

76 We study safe mapless goal-driven navigation under distribution shift. A mobile robot must reach
 77 a goal using only local observations, without access to a global map, while executing low-level
 78 velocity commands $a_t = (v_t, \omega_t)$. A navigation policy π , trained in a distribution of environments
 79 $\mathcal{E}_{\text{train}}$, proposes nominal actions $a_t^\pi = \pi(s_t)$. During deployment, the robot operates in environments
 80 $\mathcal{E}_{\text{test}}$ whose geometry and sensing or dynamics differ from training, $\mathcal{E}_{\text{test}} \not\sim \mathcal{E}_{\text{train}}$. To enforce safety,
 81 an online filter, such as CBF, is applied to project a_t^π onto a set of safe controls, ensuring forward
 82 invariance of a safe set $\mathcal{S}$ defined in Eq (3).

83 Our objective is to design a deployment-time mechanism that, given a fixed pre-trained policy π
 84 and a CBF-based safety filter, outputs control actions a_t while (i) reducing the probability of safety
 85 violations under OOD shifts, and (ii) preserving goal-reaching performance in familiar conditions
 86 by avoiding unnecessary conservatism. The mechanism must operate online, relying solely on local
 87 observations, and without requiring retraining π in the deployment environments.

88 4 Methodology

89 Figure 1 depicts the general schema of our proposed framework. We introduce an uncertainty-driven
 90 shielding mechanism that operates at deployment time on top of a fixed pre-trained mapless navigation
 91 policy. The method augments a nominal controller with a conservative safety filter, but it activates
 92 the filter only when epistemic uncertainty indicates that the current state-action pair departs from the
 93 training manifold.

94 In particular at each time step, the nominal policy proposes an action $a_t^\pi = \pi(s_t)$. The Uncertainty
 95 Filter (UF) estimates epistemic uncertainty by evaluating the ensemble disagreement of K learned
 96 one-step transition models trained on rollouts collected under π (Eq. (1)). The resulting scalar score
 97 U_t is then computed as in Eq. (2). Since the uncertainty estimator is not trained to predict barrier
 98 violations, Appendix A further analyzes the empirical relation between epistemic uncertainty and
 99 CBF-critical states. The UF subsequently gates a CBF-based safety projection where the executed
 100 action depends on whether the uncertainty estimate is below a threshold τ or exceeds it, resulting in
 101 either the nominal action or a fallback safe action, respectively. This conditional execution can be
 102 formalized as:

$$a_t = \begin{cases} a_t^\pi & \text{if } U_t \leq \tau, \\ a_t^{\text{safe}} & \text{if } U_t > \tau. \end{cases} \quad (6)$$

The threshold τ is calibrated from in-distribution uncertainty statistics computed in the training environment and is subsequently kept fixed at deployment to trigger shielding solely under detected distribution shift.

When $U_t > \tau$, the fallback action a_t^{safe} is obtained by solving the CBF quadratic program in Eq. (5), which minimally modifies the nominal command while satisfying the local barrier condition. Finally, to prevent the complete stalling typical of strict barrier constraints, our underlying CBF implementation is systematically modified to retain a small minimum forward-velocity component during any safe projection.

5 Experimental Setup

All experiments are conducted in a Unity physics simulation interfaced with our Python learning framework through the ML-Agents Toolkit [13]. We use CleanRL [14] as the base implementation for the reinforcement-learning algorithms¹.

5.1 Navigation Task

We consider a mapless goal-driven navigation task in which a mobile robot must reach a target location using only local observations and relative goal information, without access to a global map. The robot is modeled after a LIMO mobile base and is controlled through low-level velocity commands $a_t = (v_t, \omega_t)$, where v_t is the forward linear velocity and ω_t is the angular velocity.

The policy receives compact local observations with LiDAR-like ranges, robot velocity, goal geometry, and recent motion history. The agent is trained with a shaped reward that encourages progress toward the goal, penalizes stagnation, and provides a terminal bonus upon successful completion. Additional implementation details, including observation dimensionality, control frequency, and reward coefficients, are reported in Appendix B.

5.2 Environment Generation and Evaluation Scenarios

The navigation map consists of a square arena enclosed by static perimeter walls, while the interior layout is generated procedurally. The environment is discretized into a regular grid, initialized with occupied cells, and iteratively refined using a cellular automaton algorithm [15]. The generator is parameterized by

$$\Omega = (r, p, i, t_b, t_s), \quad (7)$$

where r is the grid resolution, p is the initial obstacle fill percentage, i is the number of smoothing iterations, and t_b and t_s control obstacle birth and survival.

The training environment is generated with $\Omega_{\text{train}} = (24, 0.45, 3, 6, 3)$ and uses square atomic obstacle elements. This configuration produces relatively homogeneous corridor-like structures and simple navigation paths. To evaluate robustness under distribution shift, we define two procedurally generated OOD environments by changing both the generator parameters and the obstacle geometry. In both OOD settings, cylindrical obstacles replace square wall elements, introducing local geometric patterns unseen during training.

The first OOD configuration, $\Omega_{\text{OOD1}} = (24, 0.03, 1, 1, 0)$, generates large and well-separated obstacle clusters, leading to wider paths with higher path tortuosity. The second configuration, $\Omega_{\text{OOD2}} = (30, 0.12, 1, 9, 0)$, produces smaller and more scattered obstacles, resulting in globally more direct paths that require precise local avoidance. We quantify the induced distribution shift using four global geometric scores from [15]: distance to closest obstacle, average visibility, dispersion, and characteristic dimension. The resulting metrics are summarized in Table 1.

¹Upon acceptance, we will release the code, including the implementation of environments and agents, to replicate all experiments.

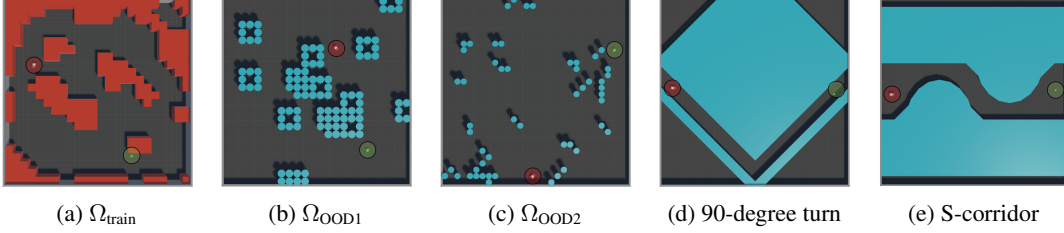


Figure 2: Evaluation environments used in the experiments. (a, b, c) Training distribution and procedurally generated OOD layouts. (d, e) Manually designed static OOD layouts isolating critical geometric configurations.

Table 1: Procedural environment metrics.

Metric	Train	OOD1	OOD2
Closest Obs.	3.71 ± 0.53	4.23 ± 0.86	3.53 ± 0.45
Visibility	11.54 ± 1.70	9.81 ± 1.57	11.26 ± 1.08
Dispersion	8.24 ± 0.99	7.28 ± 1.02	8.55 ± 0.63
Char. Dim.	11.95 ± 2.21	9.86 ± 1.79	9.94 ± 1.28

In addition to procedurally generated environments, we define two manually specified static layouts to isolate critical geometric configurations, namely a sharp 90° turn and an S-shaped corridor.

These scenarios expose the limitations of baseline approaches: unconstrained learned policies tend to collide due to overconfident execution, whereas continuously active CBF shielding often leads to stagnation.

In all environment visualizations, red and green circles mark the robot and target. Positions are randomized in procedural layouts and fixed at opposite ends in static layouts.

6 Experiments and Results

6.1 Training Details

For the proposed deployment-time safety mechanisms, the UF is trained on a dedicated dataset collected using the pre-trained policy π in the training environment Ω_{train} specifically for this purpose. The largest portion is used to train the probabilistic ensemble with a preliminary optimization stage and an elite- K selection [16], keeping only the best models for uncertainty prediction. The remaining data are used to estimate in-distribution uncertainty quantiles and choose the threshold τ .

The main evaluation focuses on SAC-based [17] policies. To assess whether the proposed gating mechanism depends on the RL backbone, supplementary PPO-based experiments are reported in Appendix C.

The SAC-based policies were trained using an ensemble of 5 critic networks. Hyperparameters were selected through a limited random search, preferred over more structured methods due to its greater efficiency in the presence of correlations between hyperparameters [18]. The SAC_p variant differs from SAC only by the inclusion of an explicit penalty term associated with collisions with walls, which biases the learned behavior toward collision avoidance already during training. LagSAC follows the standard Lagrangian constrained formulation [4] built on top of the SAC backbone and retains the same 5-critic ensemble architecture. The learning curves for all the variations are reported in Fig. 3 on a shared axis to enable direct comparison. All learning curves reach success rates above 95% at their peak performance during training.

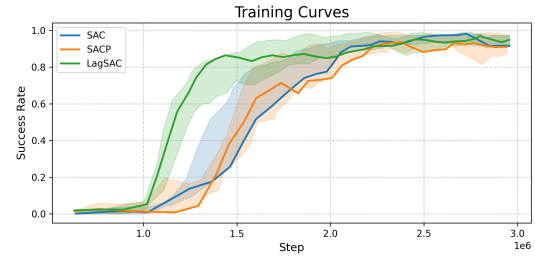


Figure 3: Policy training learning curves with shaded variance across 10 initialization seeds.

6.2 Evaluation Protocol and Metrics

We evaluate the trained policies in five test settings: two representative static OOD scenarios, two procedural OOD environments, and one real-world deployment. An episode terminates (i) upon

reaching the goal, or (ii) after a collision with an obstacle, or (iii) timeout. The episode horizon is set to 2000 physics steps, increased from the 1000 steps used during training, to limit premature timeouts and keep the analysis focused on safety and effective task completion.

For each test, we report three primary metrics: Success Rate, Collision Rate, and the average episode length computed over successful episodes, referred to as Success Episode Length (SEL). Since an episode strictly terminates with either success, collision, or timeout, the timeout rate is not explicitly reported but can be directly derived by subtracting the success and collision rates from 100%. This implicit metric is highly relevant for evaluating the stagnation tendency of safe policies. For statistical rigor, uncertainty is reported as the 95% Wilson score confidence interval for the binary metrics, success and collision rates [19, 20], and as standard deviation for the continuous SEL metric.

To ensure stable estimates under distribution shift, we evaluate each procedurally generated OOD environment over 2000 episodes, each static scenario over 250 episodes, and the real-world deployment over 25 episodes. Additionally, the uncertainty threshold τ is fixed across all deployment environments at the 0.9 quantile of the uncertainty distribution, calibrated using a separate dataset collected in the training environment.

Table 2 details the results for the main empirical evaluation. We report absolute metrics for all the configurations. Additionally for all the ablations, i.e. base policy and unfiltered vanilla CBF, we complement the table with the relative signed deviation from the UF baseline. The lagrangian variant serves as an external baseline for all the scenarios. SEL is omitted where success is 0%.

6.3 Static OOD Environments

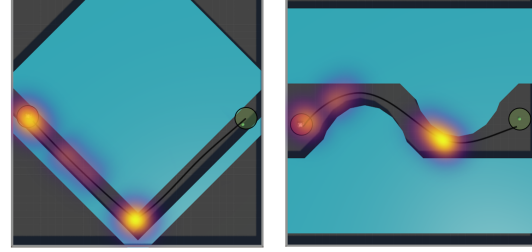
In the **90-degree turn** test (Fig. 2d), SAC+UF reaches 85.2% success with 13.2% collisions, while plain SAC drops to 10.8% success and 81.6% collisions. This confirms that the nominal policy

Table 2: SAC Results

	Configuration	Success Rate ($\uparrow$)	$\Delta_{success}$	Collision Rate ($\downarrow$)	$\Delta_{collision}$	SEL ($\downarrow$)	Δ_{SEL}
90°-turn	SAC + UF	85.2(-4.9,+3.9)		13.2(-3.6,+4.8)		995.8 $\pm$ 165.6	
	SAC	10.8(-3.3,+4.5)	-74.4	81.6(-5.3,+4.3)	+68.4	771.8 $\pm$ 91.1	-224.0
	CBF	0.0(-0.0,+1.5)	-85.2	18.4(-4.3,+5.3)	+5.2	—	—
	SACp + UF	32.0(-5.5,+6.0)		65.2(-6.1,+5.6)		1767.1 $\pm$ 316.0	
	SACp	19.2(-4.4,+5.3)	-12.8	78.8(-5.5,+4.6)	+13.6	1713.2 $\pm$ 108.5	-53.9
	CBF	0.0(-0.0,+1.5)	-32.0	20.0(-4.5,+5.4)	-45.2	—	—
	LagSAC	0.0(-0.0,+1.5)		10.4(-3.2,+4.4)		—	
S-corridor	SAC + UF	88.8(-4.5,+3.3)		8.0(-2.8,+4.0)		682.8 $\pm$ 111.5	
	SAC	68.0(-6.0,+5.5)	-20.8	32.0(-5.5,+6.0)	+24.0	627.8 $\pm$ 60.4	-55.0
	CBF	0.0(-0.0,+1.5)	-88.8	18.4(-4.3,+5.3)	+10.4	—	—
	SACp + UF	88.8(-4.5,+3.3)		5.6(-2.2,+3.6)		596.6 $\pm$ 40.4	
	SACp	90.8(-4.2,+3.0)	+2.0	7.6(-2.7,+4.0)	+2.0	585.1 $\pm$ 15.0	-11.4
	CBF	0.0(-0.0,+1.5)	-88.8	19.2(-4.4,+5.3)	+13.6	—	—
	LagSAC	90.8(-4.2,+3.0)		3.6(-1.7,+3.1)		599.9 $\pm$ 17.6	
Ω_{OOD1}	SAC + UF	84.2(-1.7,+1.5)		12.2(-1.4,+1.5)		464.4 $\pm$ 406.9	
	SAC	84.6(-1.6,+1.5)	+0.4	14.7(-1.5,+1.6)	+2.5	381.4 $\pm$ 248.2	-83.0
	CBF	81.2(-1.8,+1.7)	-3.0	9.6(-1.2,+1.4)	-2.6	582.3 $\pm$ 507.2	+117.9
	SACp + UF	85.1(-1.6,+1.5)		10.6(-1.3,+1.4)		458.1 $\pm$ 416.1	
	SACp	87.3(-1.5,+1.4)	+2.2	11.4(-1.3,+1.5)	+0.8	345.2 $\pm$ 212.4	-112.9
	CBF	80.5(-1.8,+1.7)	-4.6	7.2(-1.0,+1.2)	-3.4	557.6 $\pm$ 507.5	+99.6
	LagSAC	85.4(-1.6,+1.5)		10.1(-1.2,+1.4)		380.6 $\pm$ 273.7	
Ω_{OOD2}	SAC + UF	83.5(-1.7,+1.6)		13.8(-1.4,+1.6)		600.2 $\pm$ 464.3	
	SAC	83.2(-1.7,+1.6)	-0.2	16.8(-1.6,+1.7)	+3.0	382.5 $\pm$ 221.3	-217.7
	CBF	83.2(-1.7,+1.6)	-0.2	12.2(-1.4,+1.5)	-1.5	646.5 $\pm$ 493.2	+46.3
	SACp + UF	86.6(-1.6,+1.4)		10.8(-1.3,+1.4)		585.7 $\pm$ 462.3	
	SACp	81.8(-1.8,+1.6)	-4.8	18.1(-1.6,+1.7)	+7.2	348.8 $\pm$ 192.5	-236.9
	CBF	84.4(-1.7,+1.5)	-2.2	9.4(-1.2,+1.4)	-1.4	609.7 $\pm$ 476.5	+24.0
	LagSAC	79.4(-1.8,+1.7)		19.3(-1.7,+1.8)		385.6 $\pm$ 257.0	
Real world	SAC + UF	52.0(-18.5,+18.0)		12.0(-7.8,+18.0)		1231.3 $\pm$ 412.8	
	SAC	4.0(-3.3,+15.5)	-48.0	96.0(-15.5,+3.3)	+84.0	646.3 $\pm$ 438.1	-585.0
	CBF	32.0(-14.8,+19.6)	-20.0	32.0(-14.8,+19.6)	+20.0	1220.4 $\pm$ 533.6	-10.9
	SACp + UF	96.0(-15.5,+3.3)		4.0(-3.3,+15.5)		265.5 $\pm$ 110.3	
	SACp	0.0(-0.0,+13.3)	-96.0	100.0(-13.3,+0.0)	+96.0	—	—
	CBF	92.0(-17.0,+5.8)	-4.0	8.0(-5.8,+17.0)	+4.0	285.8 $\pm$ 113.2	+20.3
	LagSAC	0.0(-0.0,+13.3)		100.0(-13.3,+0.0)		—	

remains highly overconfident in this unseen geometry and tends to enter the corner too aggressively. Conversely, the continuously active CBF baseline reduces collisions but collapses to 0% success, as the robot becomes trapped in the narrow corner and times out. UF balances these two failure modes by preventing most catastrophic crashes while avoiding the permanent freezing induced by unconditional shielding. The increased SEL relative to SAC is consistent with more cautious navigation, rather than a loss of goal-directed behavior

For SAC_P , the improvement is weaker. A likely explanation is that this policy already learns to operate close to walls during training, so near-collision states may still appear familiar to the uncertainty estimator and fail to trigger timely intervention. Even in this case, however, UF remains the best-performing deployment-time mechanism within the SAC_P group.



(a) 90-degree turn

(b) S-shaped corridor

Figure 4: UF activation heatmaps in static environments for the SAC agent. Color intensity indicates normalized intervention frequency; black curves show mean trajectories.

The activation heatmap in Fig. 4a further clarifies this behavior. UF interventions concentrate near the inner corner, where wall proximity changes rapidly, and near the initial segment, where the robot often starts with a suboptimal heading. This indicates that corrections are not applied uniformly, but localized in geometrically critical regions poorly covered by the training distribution.

In the **S-shaped corridor**, $SAC+UF$ again improves safety over plain SAC, reducing collisions from 32.0% to 8.0% while preserving high success. Here the nominal policy is already more capable, so UF mostly acts as a lightweight safety layer with only a modest increase in SEL. By contrast, the unfiltered CBF again reaches 0% success, because continuous repulsion from the lateral walls induces oscillations or freezing along the narrow curves.

The same preservation of performance is visible for SAC_P . Since this policy already navigates the corridor reliably, UF does not substantially alter its behavior and keeps success essentially unchanged. The S-corridor heatmap in Fig. 4b shows activations mainly at the beginning of the episode and around the two bends, especially the second one, where the robot tends to cut the turn too tightly. In these regions, temporary CBF corrections slow or stop the robot long enough to recover a safer pose before continuing.

6.4 Procedural OOD Environments

In Ω_{OOD1} (Fig. 2b), the deployment-time mechanisms reveal a safety-performance trade-off. For SAC, the plain policy achieves slightly higher success and shorter SEL, but at the cost of more collisions. Continuous CBF produces the safest behavior in terms of collision rate, yet it lowers success and substantially increases SEL, showing the expected conservatism of always-on shielding. UF lies between these extremes: it keeps success close to the nominal policy while reducing collisions, with a moderate increase in SEL. Qualitatively, plain policies tend to skirt large obstacle clusters aggressively, whereas UF enforces a wider margin near corners and cluster boundaries.

The same pattern appears for SAC_P . The nominal policy is more efficient but less safe, while CBF is safer but slower and less successful. UF provides a compromise, reducing collisions relative to the plain policy without the stronger reachability loss induced by continuous shielding.

In Ω_{OOD2} (Fig. 2c), where smaller scattered obstacles create denser local bottlenecks, UF becomes more beneficial. For SAC, it preserves success while reducing collisions relative to the nominal policy. For SAC_P , it improves both success and safety, suggesting that uncertainty-triggered corrections help the robot traverse ambiguous obstacle configurations where the nominal policy is prone to unsafe shortcuts. Compared with continuous CBF, UF achieves higher success with only a small increase in

collisions and similar or lower SEL. This indicates that selective shielding improves reachability by avoiding hesitation and local trapping from continuous barriers.

6.5 Real-World Deployment

In the real-world test, the agent is deployed on a LIMO robot under a combined distribution shift: the layout is unseen, and both observations and dynamics differ from simulation due to sim-to-real mismatch.

Although the physical environment is geometrically simple (Fig. 5), the nominal SAC and SAC_P policies fail almost completely, with collision rates of 96.0% and 100.0%, respectively. This highlights the severity of the sim-to-real shift and shows that simulation-trained behaviors do not transfer reliably on their own. UF restores task completion for both policies, reaching 52.0% success for SAC+UF and 96.0% for SAC_P+UF, while keeping collisions substantially lower than the nominal policies. Since this improvement appears for both controllers, the result also suggests that the mechanism is not tied to a specific learned policy.

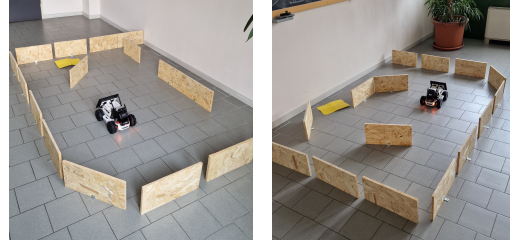


Figure 5: Different viewpoints of the physical environment used for real-world validation.

Continuous CBF also improves over the nominal baselines, but its effect is less consistent. For SAC, it recovers only part of the lost performance; for SAC_P, it approaches UF but remains slightly worse in both success and collision rate. These results suggest that the key factor in deployment is not merely adding a safety layer, but activating it selectively when the robot encounters unfamiliar observations or dynamics. The collapse of LagSAC further confirms that training-time safety constraints alone remain brittle under severe distribution shift.

These results suggest that the key requirement in real-world deployment is not simply adding a safety layer, but activating it selectively when unfamiliar observations or dynamics are detected.

7 Limitations

The proposed framework has some limitations. First, it remains dependent on the quality of the nominal policy: uncertainty-gated shielding can correct unsafe local actions, but it cannot fully recover goal-directed behavior when the policy itself fails. Second, the uncertainty threshold is calibrated offline on the training distribution and kept fixed at deployment, which may be suboptimal across different environments or sensor conditions. Third, ensemble disagreement is only a proxy for epistemic uncertainty and may not detect all risky states, especially when near-obstacle behavior is already present in the training data. Finally, the CBF layer relies on local observations and approximate dynamics, so its empirical safety benefits do not imply complete formal guarantees under severe perception errors, actuation delays, or sim-to-real mismatch.

8 Conclusions

We presented an uncertainty-gated safety framework for mapless robot navigation under distribution shift. The method uses epistemic uncertainty to selectively activate a CBF-based safety filter, enabling conservative corrections only when the robot departs from the training distribution.

Experiments across static OOD layouts, procedurally generated environments, and real-world deployment show that the proposed approach reduces unsafe behavior while avoiding the stagnation caused by continuous shielding. These results demonstrate that uncertainty-aware intervention can improve the robustness of learned navigation policies without requiring policy retraining.

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A Uncertainty as a Proxy for Collision Risk

This appendix further discusses to what extent the proposed uncertainty signal can be interpreted as an indicator of collision-related safety. This question is difficult to assess directly, since the proposed framework is broader than the collision-avoidance setting considered in this work. In the present implementation, the CBF layer encodes collision avoidance as the only explicit critical condition. By contrast, the uncertainty estimator is not trained to predict collisions or to classify violations of a specific barrier constraint, but estimates epistemic uncertainty in the learned transition model. It therefore captures a more general notion of deviation from the training distribution.

For this reason, uncertainty should not be interpreted as a direct collision detector. A collision is only one possible manifestation of unsafe or unfamiliar behavior as high uncertainty may also arise in states that are dynamically or geometrically unusual but not immediately unsafe, while some near-collision configurations may remain familiar if similar states were sufficiently represented in the training data. Nevertheless, since the safety layer used in this work is based on CBF constraints, it is meaningful to investigate whether uncertainty is informative with respect to the activation of collision-avoidance constraints.

To explore this relation, we first compare the distributions of epistemic uncertainty predicted by the ensemble model in states where the CBF constraints are satisfied and in states where they are violated. The histograms in Fig. 6 show that the two distributions are not identical, indicating that uncertainty carries information about constraint-critical states. This relation is particularly clear for the SAC-based policies: for SAC, violated-constraint states are consistently shifted toward higher uncertainty across the procedural OOD environments and the S-shaped corridor, with visible separation between the distributions. A similar, although less uniform, trend is observed for SAC_p, where violated constraints often correspond to higher uncertainty values but with stronger overlap. These results indicate that epistemic uncertainty is only partially aligned with CBF violations.

As a second analysis, we evaluate false alarms of the uncertainty-gated CBF mechanism. In this context, a false alarm occurs when the uncertainty filter activates the CBF even though the CBF constraints are not violated. While such activations do not degrade agent performance or introduce a significant computational burden, they represent unnecessary interventions with respect to the specific collision-avoidance constraint considered here. We compare these false alarms with those produced by a random filter whose activation probability is matched to the empirical activation

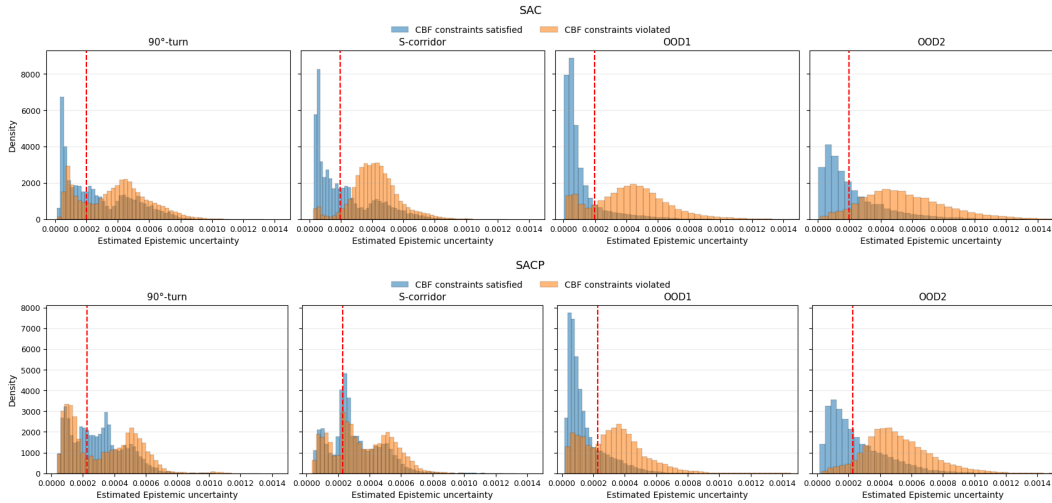


Figure 6: Distribution of predicted epistemic uncertainty for states in which the CBF constraints are satisfied or violated, across SAC-based policies and environments. The red vertical line denotes the uncertainty threshold used in the experiments, computed strictly from the training data. The distributions differ in several cases, indicating that uncertainty carries information about constraint-critical states while remaining an imperfect proxy for CBF violation.

Table 3: Comparison between the false-alarm rates of the uncertainty filter and those of a random filter with matched empirical activation probability for each SAC-based policy–environment pair. All values are reported as percentages.

Policy	Environment	UF false alarms	Random false alarms
SAC	Ω_{OOD1}	16.0 _(−0.3,+0.3)	48.8 _(−0.4,+0.4)
	Ω_{OOD2}	35.5 _(−0.3,+0.3)	53.9 _(−0.3,+0.3)
	90°-turn	11.3 _(−0.3,+0.3)	17.0 _(−0.4,+0.4)
	<i>S</i> -corridor	8.0 _(−0.2,+0.2)	15.0 _(−0.3,+0.3)
SAC _P	Ω_{OOD1}	21.0 _(−0.3,+0.3)	46.2 _(−0.4,+0.4)
	Ω_{OOD2}	34.7 _(−0.3,+0.3)	51.7 _(−0.3,+0.3)
	90°-turn	14.1 _(−0.4,+0.4)	14.6 _(−0.4,+0.4)
	<i>S</i> -corridor	14.4 _(−0.3,+0.3)	14.6 _(−0.3,+0.3)

probability of the uncertainty filter for each policy–environment pair. This baseline preserves the specific intervention rate, but removes the state-dependent information carried by uncertainty.

The results are summarized in Table 3. In general, the uncertainty-based filter produces fewer false alarms than the random filter for both SAC-based policies. The reduction is strongest for SAC, and remains clear for SAC_P. This indicates that uncertainty-triggered activations are not simply a consequence of the chosen activation probability, but are correlated with states in which CBF intervention is more likely to be relevant.

Overall, these analyses support the interpretation that the proposed uncertainty filter performs the intended role in the present collision-avoidance setting. Although epistemic uncertainty should not be interpreted as a collision detector or as a direct classifier of CBF violations, it is empirically aligned with constraint-critical states and leads to fewer unnecessary interventions than a matched random activation baseline. This indicates that the filter exploits meaningful state-dependent information rather than merely reflecting the chosen activation rate.

B Additional Experimental Details

This appendix reports additional implementation details underlying the experimental setup. While the main text focuses on the evaluation protocol and empirical results, the following sections specify the control timing, observation representation, reward design, and robot-scale assumptions used in both simulation and real-world deployment.

Control frequency. The Unity physics engine runs at 50 Hz. The policy is queried every $N = 5$ fixed physics steps, resulting in an effective policy frequency of 10 Hz. The selected velocity command (v_t, ω_t) is held constant between two consecutive policy evaluations. The forward velocity v_t is constrained to be non-negative. In contrast, the CBF layer is evaluated at every physics step, so that the safety correction can react at the simulator frequency.

Observation structure. Each policy input consists of a stack of the last four observation frames, sampled at the policy decision frequency. Each frame concatenates a LiDAR-like perception vector, implemented in Unity through 21 raycasts spanning a 180° frontal field of view, the forward and lateral components of the robot velocity, the distance and relative angle to the goal, and the distance and relative angle computed with respect to an exponential moving average of the agent’s past positions. The raycast measurements provide range-based obstacle information analogous to a planar LiDAR observation. The goal object is excluded from raycast collision detection, so sensing rays can pass through it without occluding obstacle perception. Temporal stacking provides short-term motion information without requiring recurrent architectures.

Figure 7 illustrates these quantities in a procedurally generated training environment. White rays correspond to range detection, while red rays indicate detections of nearby obstacles. The green sphere denotes the goal. The blue region shows the area within which the EMA-based stagnation penalty is active, whereas the red region around the robot visualizes the local CBF safety area.

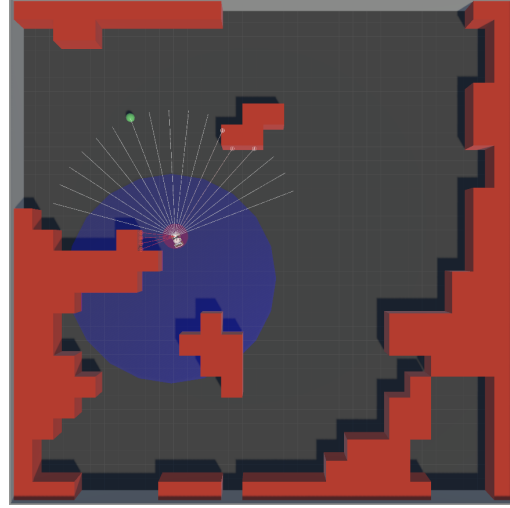


Figure 7: Debug visualization of raycast-based perception, EMA penalty region, and local CBF safety area.

Reward shaping. The reward provided to the agent at each step is composed of progress toward the goal, a stagnation penalty, and a terminal success bonus:

$$r_t = w_p \tanh \left(\frac{\max(0, d_{t-1} - d_t)}{v_{\max} \Delta t + \epsilon} \right) - w_s \left[\text{clip}_{[0,1]} \left(\frac{R - \|p_t - \hat{p}_t\|}{R} \right) \right]^2 + w_g \mathbb{I}_{\text{goal}}, \quad (8)$$

where d_t is the Euclidean distance to the goal, $v_{\max}$ is the maximum forward speed, Δt is the control timestep, ϵ is a small numerical constant, p_t is the current position, $\hat{p}_t$ is an exponential moving average of past positions, and R defines the range within which lack of displacement is penalized. The stagnation term reaches its maximum penalty when the robot remains close to its recent average position and vanishes once the displacement from this average exceeds R .

Robot scale. The simulated robot is built to match the physical scale of the LIMO platform. Consequently, the robot may appear relatively small in the environment visualizations, especially in the top-down renderings of the procedurally generated maps.

C PPO-Based Policies

This appendix evaluates whether the proposed uncertainty-gated safety mechanism generalizes beyond the SAC-based policies considered in the main experiments. To this end, we repeat the evaluation using PPO-based navigation policies trained under the same environment distribution and tested under the same OOD scenarios. The goal of this additional analysis is not to introduce a new algorithmic variant, but to assess whether the benefits of uncertainty-gated shielding depend on the specific RL backbone or instead arise from the deployment-time gating mechanism itself.

The PPO policy follows the standard reference implementation [21]. Similarly to the SAC family, the PPO_p variant differs from PPO only by the inclusion of an explicit collision penalty during training, while LagPPO serves as the corresponding Lagrangian constrained baseline. The corresponding training curves are reported in Fig. 8. The PPO-based results are reported in Table 4.

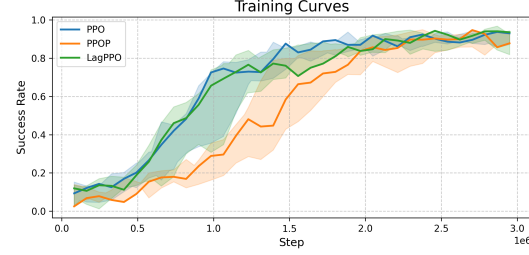


Figure 8: PPO policy training learning curves with shaded variance across 10 initialization seeds.

In the **90-degree turn**, the unconstrained PPO and PPO_p variants exhibit a near-total failure to reach the target, achieving a success rate of effectively zero. Qualitatively, these agents tend to collide with the walls halfway through the first straight segment, as soon as the right-angled corner enters their perception. This is reflected quantitatively by collision rates of approximately 100.0% and 98.4%, respectively. Despite the poor performance caused by the pre-trained policy, CBF-based shielding prevents most collisions, converting most failures into stagnation, yielding near 75% timeout rates. Our method remains robust also in this scenario mimicking the behavior of the CBF variations, highlighting a correct activation of the UF.

In the **S-shaped corridor**, the nominal policies already exhibit high proficiency in navigating the tight curves. The selective application of safety constraints via UF preserves the baseline capabilities with only negligible degradation. Conversely, the always-active CBF variants show a substantial degradation in success rate. Since the gating mechanism correctly recognizes these states as familiar, it refrains from unnecessary interventions. As a result, the uncertainty-driven approach maintains consistently high success rates across these algorithms, demonstrating that our safety layer imposes minimal performance overhead on an already capable policy.

In Ω_{OOD1} and Ω_{OOD2} , the monotonic trade-off observed for the SAC family does not fully apply to PPO-based policies. For both PPO and PPO_p, the UF-augmented versions improve over their nominal counterparts by simultaneously increasing success rates and reducing collisions. This effect is especially relevant in Ω_{OOD2} , where denser local bottlenecks make nominal policies more prone to unsafe shortcuts. Compared with continuous CBF shielding, UF achieves higher task completion while avoiding the substantial SEL increase and local stagnation often induced by always-active corrections, with only marginally higher collision rates. Overall, uncertainty-gated interventions preserve the main safety benefits of deployment-time correction without imposing the excessive conservatism of unfiltered CBFs.

In the real-world deployment, the nominal PPO and PPO_p policies fail almost completely when transferred to the physical platform, showing that behaviors learned in simulation do not generalize reliably on their own even in a geometrically simple environment. Introducing uncertainty-gated shielding consistently restores task completion while keeping collision rates substantially lower. Continuous CBF shielding also improves over the nominal policies, but its effect is less consistent: depending on the underlying controller, it either recovers only part of the lost performance or becomes overly conservative and interferes with progress. The collapse of the Lagrangian baseline further reinforces that training-time safety constraints remain brittle under severe distribution shift, while uncertainty-gated corrections offer a more reliable deployment-time alternative.

Table 4: PPO Results

	Configuration	Success Rate ($\uparrow$)	$\Delta_{success}$	Collision Rate ($\downarrow$)	$\Delta_{collision}$	SEL ($\downarrow$)	Δ_{SEL}
90°-turn	PPO + UF	0.0 _(-0.0,+0.8)		20.0 _(-3.3,+3.7)		—	
	PPO	0.0 _(-0.0,+0.8)	+0.0	99.0 _(-1.3,+0.6)	+79.0	—	—
	PPO + CBF	0.2 _(-0.2,+0.9)	+0.2	25.2 _(-3.6,+4.0)	+5.2	242.5 $\pm$ 236.9	—
	PPO _p + UF	0.0 _(-0.0,+0.8)		26.8 _(-3.7,+4.0)		—	
	PPO _p	0.4 _(-0.4,+1.0)	+0.4	98.6 _(-1.5,+0.7)	+71.8	1053.4 $\pm$ 282.0	—
	PPO _p + CBF	0.0 _(-0.0,+0.8)	+0.0	24.6 _(-3.6,+4.0)	-2.2	—	—
	LAGPPO	28.4 _(-3.8,+4.1)		69.2 _(-4.2,+3.9)		934.8 $\pm$ 202.1	
S-corridor	PPO + UF	86.6 _(-3.3,+2.7)		12.4 _(-2.6,+3.2)		643.2 $\pm$ 77.9	
	PPO	86.8 _(-3.2,+2.7)	+0.2	11.6 _(-2.5,+3.1)	-0.8	628.4 $\pm$ 56.9	-14.7
	PPO + CBF	14.4 _(-2.8,+3.3)	-72.2	17.6 _(-3.1,+3.6)	+5.2	1864.5 $\pm$ 228.4	+1221.3
	PPO _p + UF	88.6 _(-3.1,+2.5)		10.6 _(-2.4,+3.0)		706.1 $\pm$ 103.2	
	PPO _p	87.2 _(-3.2,+2.6)	-1.4	9.2 _(-2.2,+2.9)	-1.4	691.6 $\pm$ 39.5	-14.4
	PPO _p + CBF	7.0 _(-1.9,+2.6)	-81.6	18.2 _(-3.1,+3.6)	+7.6	1712.5 $\pm$ 447.7	+1006.4
	LAGPPO	57.2 _(-4.4,+4.3)		39.2 _(-4.2,+4.3)		751.6 $\pm$ 52.7	
Ω_{OOD1}	PPO + UF	79.7 _(-1.8,+1.7)		19.6 _(-1.7,+1.8)		501.6 $\pm$ 398.6	
	PPO	74.6 _(-2.0,+1.9)	-5.1	25.4 _(-1.9,+2.0)	+5.8	402.2 $\pm$ 228.6	-99.4
	PPO + CBF	76.5 _(-1.9,+1.8)	-3.2	16.8 _(-1.6,+1.7)	-2.9	590.6 $\pm$ 491.5	+89.0
	PPO _p + UF	78.3 _(-1.9,+1.7)		21.3 _(-1.7,+1.8)		495.3 $\pm$ 357.4	
	PPO _p	74.0 _(-2.0,+1.9)	-4.3	25.9 _(-1.9,+2.0)	+4.6	452.6 $\pm$ 270.7	-42.6
	PPO _p + CBF	73.0 _(-2.0,+1.9)	-5.4	16.3 _(-1.6,+1.7)	-5.0	611.9 $\pm$ 501.9	+116.6
	LAGPPO	81.5 _(-1.8,+1.6)		18.6 _(-1.6,+1.8)		534.4 $\pm$ 274.0	
Ω_{OOD2}	PPO + UF	80.0 _(-1.8,+1.7)		19.5 _(-1.7,+1.8)		592.4 $\pm$ 432.0	
	PPO	75.6 _(-1.9,+1.8)	-4.4	24.4 _(-1.8,+1.9)	+4.9	413.4 $\pm$ 240.4	-179.0
	PPO + CBF	77.1 _(-1.9,+1.8)	-2.9	18.6 _(-1.6,+1.8)	-0.9	663.8 $\pm$ 490.4	+71.4
	PPO _p + UF	78.6 _(-1.8,+1.7)		21.2 _(-1.7,+1.8)		512.8 $\pm$ 325.3	
	PPO _p	75.8 _(-1.9,+1.8)	-2.8	24.2 _(-1.8,+1.9)	+2.9	454.4 $\pm$ 266.5	-58.3
	PPO _p + CBF	73.8 _(-2.0,+1.9)	-4.9	19.3 _(-1.7,+1.8)	-1.9	705.7 $\pm$ 516.1	+192.9
	LAGPPO	82.7 _(-1.7,+1.6)		17.3 _(-1.6,+1.7)		559.4 $\pm$ 274.0	
Real world	PPO + UF	72.0 _(-9.5,+7.9)		28.0 _(-7.9,+9.5)		581.5 $\pm$ 352.3	
	PPO	0.0 _(-0.0,+3.7)	-72.0	100.0 _(-3.7,+0.0)	+72.0	—	—
	PPO + CBF	48.0 _(-9.5,+9.7)	-24.0	52.0 _(-9.7,+9.5)	+24.0	560.3 $\pm$ 304.4	-21.2
	PPO _p + UF	68.0 _(-9.7,+8.3)		32.0 _(-8.3,+9.7)		487.5 $\pm$ 170.6	
	PPO _p	0.0 _(-0.0,+3.7)	-68.0	100.0 _(-3.7,+0.0)	+68.0	—	—
	PPO _p + CBF	20.0 _(-6.7,+8.9)	-48.0	80.0 _(-8.9,+6.7)	+48.0	832.1 $\pm$ 488.2	+344.6
	LAGPPO	0.0 _(-0.0,+3.7)		100.0 _(-3.7,+0.0)		—	